

FATHIMA SHIFA MACHINGAL

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EDUCATION

South Dakota School of Mines and Technology , Rapid City, SD Advisor: Prof. Matthias Plum	August 2025 – Present
Indian Institute of Technology Hyderabad , India <i>M.Sc. in Physics</i> Master's Thesis: Search for spatial coincidences between IceCube neutrinos and galactic sources Advisor: Prof. Shantanu Desai	August 2023 – May 2025
University of Calicut , Kerala, India <i>B.Sc. in Physics</i>	August 2020 – May 2023

PEER-REVIEWED PUBLICATIONS

- 2 first-author publications in high-impact journals (*JHEAP*, *Phys. Dark Universe*)
1. **F. Machingal** & S. Desai, "Search for spatial coincidence between magnetars and IceCube detected neutrinos", *Phys. Dark Universe* (2026). [[PDU](#)]
 2. **F. Machingal** & S. Desai, "Search for spatial coincidence between IceCube neutrinos and gamma-ray bright red dwarfs", *J. High Energy Astrophys.* (2025). [[JHEAP](#)]

AWARDS & HONORS

Rose Weinstein & Olga Pnitch (MIFF 08) Fellowship	2025
National Scholarship for Postgraduate Studies, India	2025
IIT JAM Qualified (AIR 736)	2023
Centre Topper in National Graduate Physics Examination	2023
JEE Mains Qualified (97th percentile)	2020

PROJECT AND RESEARCH EXPERIENCE

Search for spatial coincidences between IceCube neutrinos and galactic sources Supervisor: Prof. Shantanu Desai, IIT Hyderabad, India	July 2024 – April 2025
Evaluated the spatial coincidence of IceCube neutrinos with TeV-bright red dwarfs, magnetars, and Fast Radio Bursts (FRBs).	
Non-Thermal Emission from Blazars Supervisor: Dr. Sulfikkarali NK, Farook College, Kerala, India	June 2022 – April 2023
- Analyzed the spectral energy distribution of active galactic nuclei, specifically focusing on Blazars. - Derived an approximate analytical solution for synchrotron emission and successfully fitted it to observational data gathered from Mkn 421.	

PRESENTATIONS

<i>IceCube Master Class: Moon Shadow Analysis</i> Talk delivered at IceCube Master Class, SD Mines	April 2026
<i>Calibration of 61-pixel IceAct Camera using Muon Charge</i> Presented at 16th Annual Student Research Symposium & Seminar, SD Mines	February – March 2026

Search for Spatial Coincidence between IceCube Neutrinos and Gamma-ray Bright Red Dwarfs October 2025
Seminar presentation at SD Mines

Astroparticle Physics July 2025
Invited Talk for Undergraduate students at Farook College, India

WORKSHOPS & SUMMER SCHOOLS

- International School of Cosmic Ray Astrophysics (Selected) July 2026
- IceCube Summer School 2026 (Selected) June 2026
- School on Experimental Astroparticle Physics, Ooty, India March 2025
- IIST Astronomy and Astrophysics Summer School, Kerala, India July 2024
- Beginning Astronomy v3: Data-Driven Journey Workshop, IIT Hyderabad December 2023

TEACHING & MENTORING EXPERIENCE

Teaching Assistant, South Dakota School of Mines and Technology, USA

- Computational Physics Lab January – May 2026
- Physics Lab November 2025 – May 2026
- Modern Physics November – December 2025

TECHNICAL SKILLS

Programming & Simulation: Python, Geant4, Git

Statistical Methods: Maximum Likelihood Estimation, Data Fitting

SERVICE AND OUTREACH

- Member, FIRST (Forum for Innovation and Research in Science and Technology) April 2023 – Present
- Mentor, Sunshine Counselling at IIT Hyderabad August 2024 – April 2025
- Member, Nakshatra Astronomy Club, IIT Hyderabad March 2023 – April 2025